The Answers

Note: the numbers inside the bracket in the answer are part mark in that question

1.

a)

i) What are the defining parameters in modelling a lightning surge in power networks?

[2]

Ans: large current in kA [1 mark] , steep rise (micro second) and slower fall time [1 mark]. Typically 1.2micro-seconds/50 microseconds

ii) What components design does the lightning surge influence?

[2]

Ans: lightning protection conductor [1 mark] in the overhead and tower footing resistance [1 mark].

iii) Why are usually the stator windings of a synchronous generator star connected?

[2]

Ans: I am expecting the student to approach with enough passage for earth fault current [1 mark] through neutral grounding [1 mark] for earth fault protection

b)

i) Briefly narrate the nature and importance of transient or first swing stability in the context of power system operation.

[4]

Ans: I am expecting the students to focus on the size and pace of the appearance of the fault [1 mark] , then they should focus on how it results in unbalanced between mechanical input and out power , how that will lead to rapid change (both way) of the rotor speed [1 mark] . This will then build an argument for loss of synchronism – requirement of rapidly appearing synchronising torque [1 mark]. Beyond the stability of the generator – the student then can indicate the stability of the rest of the system as it impacts the system as well …it is the first few seconds of the rotor behaviour and loss of synchronisation as matter of importance [1 mark]

ii) What are the modelling considerations for analysing transient stability problem in power systems?

[4]

Ans: The student should approach with mechnical swing being the primary concern [1 mark]– should focus on rotor circuit dyanmics [1 mark], can neglect stator trainsients which are of power frequency , so treat them as algebraic [1 mark]. Swing equations (speed and angle).. may or may not include rotor electrical damping effect – but most imporatnly mentioning of nonlinear differential and algebraic sets of equations [1 mark]

iii) How are these modelling considerations influenced by the increasing presence of renewable generation in the grid?

[4]

Ans: The student is expected to start with the difference between synchronous and asynchronous technology [1 mark] . I am looking for an answer how power converter is an decoupling device and diminishes the loss of synchronism [1 mark] . Then they would focus on control interaction between converters through networks [1 mark] – which will require important consideration. The frequency will shift from 1-2 Hz to higher value – so a change in modelling requirement – not DAE but all differential equation – EMT framework [1 mark].

c)

i) What is the purpose of speed governing in a grid-connected generator?

[3]

Ans: There is always random fluctuation in demand as it is open to customer to switch on switch off [1 mark]. The network is an AC system which requires maintaining frequency – so any random fluctuation or small changes will result in frequency change from nominal [1 mark] . This has to be adjusted by valve/gate allowing the fuel input used for mechanical torque production .. governing is a mechanism needed to main the speed of a generator at synchronous speed and hence system frequency – so student should relate the speed governing as primary frequency control means.[1 mark]

ii) Why is it relatively difficult to design the speed governing system for a hydro turbine coupled with a large synchronous generator?

[4]

Ans: The answer should start with hydro turbine dynamics – with non-minimal phase zero [2 marks] – thus allowing less phase margin – with high gain in governor control – it may lead to unstable operation of turbine [1 mark] – so very fast turbine torque control is not possible – to start with there should a transient gain reduction block [1 mark] – which during first 1-2 seconds will not let the generator to loose synchronism or have negative stability margin.

[25 marks]

2.

a)

i) Why is the rotor of a synchronous generator made of solid construction whereas its stator is laminated?

[3]

Ans: I am assessing the student's understanding that during steady state there is no emf induced in rotor body – so no circulating current. Also I am expecting during the deviation of rotor speed from synchronous value there is damping effect offered by the solid rotor. They need to mention the role of rotor body as damper during post fault clearance and recovery statge [2 marks]. The stator on the other hand carries power frequency current – so there is iron losses (hysteresis and eddy current).. lamination is the way to minimise the iron losses in the stator [1 mark]

ii) What is the purpose of closed-circuit amortisseur windings in synchronous machine rotor?

[3]

Ans: I am expecting them to write the existence(non existence) of current during transient (steady) state operation – because of the absence of current during steady state keeping shorted does not lead to additional copper losses [1 mark] . When rotor speed is different from synchronous spped there will be current , losses and temperature rise of rotor – but that temporary in nature and desirable from transient and small signal stability perspective [2 marks].

iii) Why such windings are not present in wind turbine rotor?

[3]

Ans: wind turbine rotor carries AC currents – having closed winding will lead to loss of power/torque and efficiency of power conversion [2 marks] . Also electrical damping in wind turbine is not must – as the nature of the oscillatory instability in converter interfaced wind generator is primarily electrical in nature which can be addressed through additional converter control [1 mark].

iv) How are amortisseur windings modelled in power system studies?

[3]

Ans: for stability studies they are treated as closed electrical circuit with no impressed voltage [1 mark]. More than one equivalent circulit is conceptualised to represent the whole rotor body (outer and inner) [1 mark] . They should all be differential equations [1 mark] . During steady state there is no current in them – so they are not included.

v) A round rotor synchronous generator has a synchronous reactance of 2.5 pu on its own base. When a sudden short circuit appears at the terminal with rated pre-fault voltage – the initial current is 4-5 times the rated current. It lasts for about 5 seconds before falling gradually. Why is the generator initial output current so high?

[4]

Ans: The students should demonstrate their understanding that during fault synchronous reactance does not exist [1 mark]

They then mention about the electromagnetic flux linkage geometry during fault – sub transients, transients etc and how the varying flux path lead to variable inductance with time .. how that lead to larger currents etc. [3 marks]

vi) *A synchronous generator is operating at full load with rated excitation. For some reason excitation system monitoring sends an alert; the level of excitation is reduced by 10%. How will it affect the transient stability of the generator?*

[3]

Ans: when excitation voltage is reduced by 10%, the synchronising strength drops – the peak value of the power angle curve drops [1 mark]. To deliver fixed amount of power machine load angle will rise [1 mark]. This will lead to less margin of stability [1 mark].

vii) *What is the purpose of d-q transformation in synchronous machine modelling and analysis?*

[3]

Ans: the mutual inductance between phase winding of stator and between stator and rotor winding are function of effective airgap in the magnetic flux path [1 mark] . The field axis of the rotor with respect to stator is dependent on load delivered , thus a unique inductance can not be defined irrespective of loading [1 mark]. This is overcome by d-q transformation by resolving the magnetic effect of stator coils by two coils quadrature to each other – one is aligned with filed axis (d-axis) and another along q-aixs – the reference frame of these two coils are synchronously rotating.[1 mark]

viii) *Two synchronous generators of identical capacity are delivering full load to the grid. One of them has uniform air gap between stator and rotor and the other does not. For some reason, the loss of excitation relay operates in both. Which generator must have faster and stronger control to prevent going out of synchronism and why?*

[1+2]

Ans: The student should say the one with uniform airgap will need stronger and faster control [1 mark] . The power angle relationship of salinet and nonsalient pole generator can be referred to and linked with reluctance power or torque [2 marks]

[25 marks]

3.

a)

Fig Q3.1 shows a simple power system model with one synchronous generator connected to a power network through a generator transformer and a section of power transmission line to deliver real and reactive power to the system. The values of various operating variables and parameters shown in the diagram are as follows.
E = 3.2 pu, Vt = 1.0 p.u, Xs= 2.7 p.u, Xt =0.08 p.u, Xl= 0.15 p.u , δ = 30 degree.

i) Ignoring stator resistance, derive the power angle equations for both active and reactive power from the first principle.

[4]

ii) With the help of the derived power angle equations obtained in 3 a) (i) compute the real and reactive power at the network end.

[2]

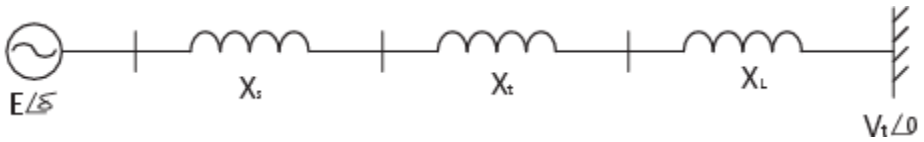


Fig Q3.1 A Simple Power System

(i) Ans

$X = X_s + X_t + X_L = 2.7 + 0.08 + 0.15 = 2.93\text{pu}$

Complex power received at the network end: $V_t I^*$

$$I = \frac{E\langle\delta - V_t\langle 0}{jX} \Rightarrow I^* = \frac{E\langle -\delta - V_t\langle 0}{-jX} = \frac{E\langle -\delta - V_t\langle 0}{X}\langle \pi/2 = \frac{E\langle(\pi/2-\delta) - V_t\langle \pi/2}{X}$$

$$S = P + jQ = V_t\langle 0 \frac{E\langle(\pi/2-\delta) - V_t\langle \pi/2}{X} = \frac{EV_t}{X}\sin\delta + j\left[\frac{EV_t\cos\delta - V_t^2}{X}\right]$$ [4 marks]

(ii)

$$P = \frac{EV_t}{X}\sin\delta = \frac{3.2\times1}{2.93}\sin 3\,0 = \frac{3.2}{2.93} \approx 0.546\text{pu}$$

$$Q = \frac{EV_t\cos\delta - V_t^2}{X} = \frac{3.2\times1\times\cos 30 - 1^2}{2.93} = \frac{1.6}{2.3} \approx 0.6045\text{pu}$$ [2 marks]

b)

Because of a transient disturbance in the system, the network voltage dropped slightly and triggering the electromechanical dynamics of the generator connected to the system. Assuming a classical swing equation model (speed ω_r and load angle δ as only dynamic variables) with an inertia constant H (sec) and damping constant D (in pu);

i) Write down the swing equations.

[2]

Ans:

$$\frac{d\delta}{dt} = \omega - \omega_s$$

$$\frac{d\omega}{dt} = \frac{\omega_s}{2H}\left[\left(P_m - \frac{EV_t}{X_s+X_t+X_l}\sin\delta - \frac{D}{\omega_s}(\omega-\omega_s)\right)\right]$$ [2 marks]

ii) *For the values of the operational variables listed in 3(a) and constants listed below, obtain the linearized dynamics in state-space form treating mechanical power as input and generator real power as output.*

H = 5.0 sec, D = 10.0 pu, operating frequency is 50 Hz or 314 rad/sec. [4]

Ans:

$$\frac{d\Delta\delta}{dt} = \Delta\omega$$

$$\frac{d\Delta\omega}{dt} = \frac{\omega_s}{2H}\left[\left(\Delta P_m - \frac{E_0V_t}{X_s+X_L+X_t}\cos\delta_0\Delta\delta - \frac{D}{\omega_s}\Delta\omega\right)\right]$$

[2marks]

In state space form,

$$\begin{bmatrix}\Delta\dot{\delta}\\ \Delta\dot{\omega}\end{bmatrix} = \begin{bmatrix}0 & 1\\ -\frac{\omega_s}{2H}\frac{E_0V_t}{X_s+X_L+X_t}\cos\delta_0 & -\frac{D}{2H}\end{bmatrix}\begin{bmatrix}\Delta\delta\\ \Delta\omega\end{bmatrix} + \begin{bmatrix}0\\ \frac{\omega_s}{2H}\end{bmatrix}[\Delta P_m]$$

[2marks]

(iii) *With the help of the states, input and output matrix obtain the transfer function polynomial in 's' with all numerical coefficients.*

[8]

First obtain the element of state matrix with the values ; [2 marks]

A = [0,1;-29.7,-1.0] Input matrix : B =[0; 31.4], Output matrix: C =[$\frac{E_0V_t}{X_s+X_L+X_t}\cos\delta_0$, 0] [2 marks]

Now find the transfer function = $C(sI-A)^{-1}B = \frac{-882.04}{S^2+S+29.7}$

[4 marks]

iv) Comment on the small-signal stability of the system with the help of the roots of the denominator polynomial obtained in 3 b) (iii).

[5]

Ans: The roots of denominator polynomial = -0.5+j5.4267; -0.5-j5.4267. [2 marks]

The system has complex conjugate poles – so it is oscillatory in nature, since the real part or the poles are in the left half of eigen-plane, it is oscillatory stable . [3 marks]

[25 marks]

4.

a)

i) List two popular wind power generation technologies out of all possible ones that have been tried in the wind power industry.

[2]

Ans: I am expecting them to mention Type 3 (DFIG) [1 mark] and Type 4 (FCIG) [1 mark]

ii) From the perspective of stability of operation, what will be the features for a design and control engineer to focus on a popular technology of generation?

[4]

Ans: I am expecting them to stress on the controllability [1 mark] and degree of freedom [1 mark]. They need to mention how reactive power [1 mark] , fault ride through [1 mark] , speed control in subrated speed (MPPT) can be controlled through grid and machine side converter control offering four independent variables to be controlled [1 mark]

iii) Which of the transient and small-signal stability will be more important for a typically sized wind turbine generator and why?

[1+3]

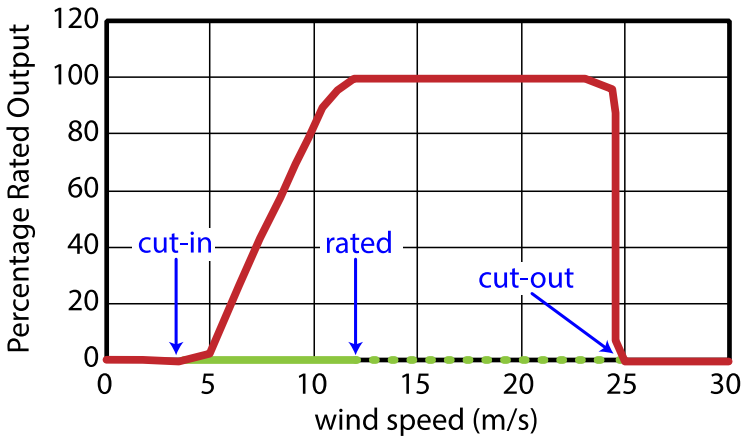
Ans: I am expecting them to suggest small signal stability as converter decouples the grid connected WTG into two systems [1 mark]. I am expecting them to touch on control instability, phase lock loop instability , low frequency oscillation in network voltage and reactive power flow . I would expect them to term as converter control driven small signal stability issue. They should mention these are new form of network instability [3 marks].

b)

ii) With the help of power output v/s wind speed define, cut-in, rated and cut-out wind speed of a wind turbine generator (WTG)

[3]

Ans: they can draw a simple power v/s speed like the following and mark the speed range



(3)

ii) Mention traditional methods of WTG control in all speed range of operation.

[3]

Ans: below cutin and beyond cutout speed the turbine is locked. [1 mark]
Between cutin and rated speed electrical control in the form of increased torque with current limits is execercised . This can be done through converter outloop control [1 mark]
Between rated and cutout speed I am expecting them to mention control on the mechanical side such as pitch control with power at rated value [1 mark]

iii) What is maximum power point tracking in relation to the operation of a WTG?

[2]

Ans: Between cutin speed and rated speed – the wind power output is nonlinear function of wind speed. In this region it is important to capture as much power is available. It will not be rated but less The MPPT works through rotor current control to produce torque that helps operate the machine as optimal tip speed ratio by controlling the rotor speed. [2 marks]

c)

i) What is the performance coefficient (C_p) of a wind turbine ?

[2]

Ans*:* The performance co-efficient is the conversion factor in wind energy production. For a particular wind turbine design it is the fraction of the incoming kinetic energy of the wind that is converted to mechanical through rotation of turbine blade. It is function of incoming wind speed and the speed of the wind immediately after leaving the turbine blade. [2 marks]

ii) What are the factors influencing the performance coefficient (C_p)?

[2]

Ans: I am expecting them to answer , blade shape so pitch angle , tip speed ratio mainly [2 marks]

iii) What maximum percentage of power from the wind can be converted to mechanical form ?

[1]

Ans 59.3%

[1 mark]

(iv) What is the significance of *wake* in the context of wind farm operation ?

[2]

Ans: I am expecting them to first describe what wake is , then how that is seen by subsequent rows of turbine ; how slowd down wind results in reduced output from trailing rows in the farm . In summary they are expected to answer overall reduction in power output and non-uniform output from the turbines in the farm

[2 marks]

[25 marks]